

Accessible Design for Diabetic Foot Ulcer Self-Monitoring: Insights into mHealth Application Design

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ABSTRACT

Individuals worldwide are living with diabetes, and under-resourced communities are more at risk of complications such as diabetic foot ulcers (DFUs). Integrating technology for DFU management is effective, but a digital gap persists especially for older adults. Older adults exhibit unfamiliarity with application design which may act as a deterrent from incorporating technology into their health self-management. To address this challenge, we conducted cognitive walkthroughs and semi-structured interviews with older adults (N=12) from under-resourced communities. We sought to understand their ability to use an app that required taking images and videos, interacting with the phone's motor/vibration system, and completing a walking test. In order to explore whether comprehension and perceived utility are due to a lapse in older adult technology understanding or flawed application design, we conducted a second study with tech savvy university students (N=10). While the students were able to successfully complete all of the walkthrough tasks, the utility of the app was limited by the diabetes knowledge of the participant groups. We emphasize the importance of community-based research initiatives and of designing mobile health technologies that are useful and usable to those who face the greatest disease burden.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods.**

KEYWORDS

Type 2 Diabetes, Mobile Health, Technology Literacy, Older Adults, Under-Resourced Communities

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1 INTRODUCTION

Although healthcare is an essential domain, its heavy reliance on technology poses challenges for older adults, especially in managing chronic conditions such as diabetes [33]. The digital divide affects marginalized groups from under-resourced communities who lack access and familiarity with these technologies [22]. Although older adults demonstrate difficulty in utilizing technology, mobile health (mHealth) interventions are known to bolster their self-confidence [34] and interest in learning about their health [34]. Type 2 Diabetes leads to various complications such as diabetic foot ulcers (DFUs). Risk factors for DFUs, lesions on the foot, include neuropathy, arterial disease, and repeated trauma, ulcers, or amputations [10, 11]. Given the nature of this potentially severe condition, diabetic foot care technology has been developed ranging from sensing systems that mitigate the risk of ulcer development [35] to wearable technologies, such as smart sole systems [26].

Over 400 million people worldwide have diabetes [27] and up to 34% of those with Type 2 Diabetes may develop DFUs [3]. Of those who develop a DFU, a fifth will require amputation with a mortality rate of up to 70% within 5 years [3]. Older adults and communities with lower-resources are disproportionately impacted by diabetes complications [14]. Early detection of symptom development is crucial in mitigating further foot complications [1].

In support of more inclusive and accessible mobile health (mHealth) application design, we evaluated a mobile application for DFU self-monitoring. Findings indicated that older adults had difficulty with the app. Rather than simply assuming this was due to their inadequacy in using technology, we decided to test the design of the application with tech savvy university students. We provide two key contributions:

- Findings related to the usability challenges and design preferences of a diabetic foot self-monitoring mHealth application by both older adults and university students
- Design implications to bolster the accessibility and usefulness of a mHealth application for older adults in low-income communities

2 RELATED WORK

2.1 Older Adults & Technology

There is a growing concern that older adults, especially those from lower-income communities, may fall behind in technology-driven societies if they are not educated about the latest technologies [4]. More than 60% of people over 50 feel that their needs are not adequately addressed by current Information and Communication Technologies [2]. However, contrasting findings suggest that lower-income individuals may exhibit less interest in adopting healthcare

technologies, exacerbating the digital gap [4]. This observation is particularly relevant as this study focuses on a lower-income older adult population.

User-centered design has proven its value by enabling HCI researchers and designers to create more useful and usable products for their intended audience [19, 30]. The socioeconomic impact of the digital divide is an important factor for user-centered design: the divide may reinforce education through wealth [13]. Researchers have also emphasized the importance of participatory design with older adults, as their voice is often not taken into consideration [24]. We add to the literature on designing for older adults by exploring the influence of both technology understanding and application design on the digital divide. Further, we also contribute to community-based research as we interacted with these participants in their living facility or place of worship.

2.2 Mobile Health Applications for DFU Care

Ubiquitous computing research has shown that pervasive technology can encourage people to engage in healthy living [15]. Mobile health applications (mHealth apps) in particular may support diabetes prevention and management by empowering individuals with information [12]. The majority of individuals living with diabetes have smart phones and are inclined to use them for diabetes management [29]. In healthcare settings, mHealth interventions have been shown to contribute to time management and cost reduction, and facilitate better treatment at point of care [31]. Medical professionals have also shown interest in mHealth [23], however, there are concerns about patients' education on diabetic foot self-management and if they would use these apps successfully [35].

A number of mHealth apps have previously been developed for diabetic foot care [18]. "FootSnap" is a clinician-facing app designed to detect the development of DFUs in a hospital setting [40]. Patient-facing interventions include "Diabetic Foot Smart App," which focuses on diabetes education [41], and "Intellin," which allows users to track their diabetes symptoms [18]. Both of these apps have not specifically been tested in the older adult and lower-socioeconomic status (SES) space, and excluded those with mental or physical impairments, such as blindness [18, 41]. Keegan et al developed and evaluated a DFU self-monitoring application with older adults, however their participants were from a less socially disadvantaged neighborhood [20]. We contribute to ongoing research in mHealth for diabetic foot care by examining how older adults with lower socioeconomic status (SES) understand and use a novel mHealth foot self-monitoring application compared to university students at a STEM aligned university. Our work supports mHealth design which is user-friendly, inclusive, and accessible to users of varying backgrounds.

3 APPLICATION DESIGN

Our mobile application prompts patients to collect data on four aspects of foot health as outlined in the Comprehensive Foot Exam for diabetes [5]. It is a patient-facing mobile application which aims to increase patient engagement and support self-advocacy at point of care. The research team first developed a low-fidelity prototype and then iterated on it based on participant feedback. The dermatological update (image capture) instructs the user to take a photo

of their foot to capture visual indications of wounds or infection. The capillary refill test (video capture) measures how fast the toe will regain its color after pressure is applied, to assess microcirculation [21]. The third test is the neurological test (vibration test), which plays a 10-second vibration while the corner of the phone is placed on the big toe, to detect loss of sensation. Finally, during the walking test, the user walks for two minutes while they hold the phone. The captured gait data measures how the user is walking and may signal musculoskeletal abnormalities, irregular circulation, or issues developing in the lower extremities [6].

Users can view their previous recordings on the "Activity Page." They can select and add a new record (image capture, video capture, vibration test, or walking test) through the "Add a Record Page." The final page in the app is "My Account," which has account details, security (password change), report a bug, and request help. These three pages are accessed through the clipboard, plus, and profile icons, respectively (See Figure 1).

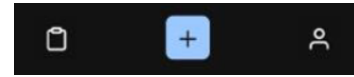


Figure 1: The app's three icons: clipboard (left), plus (middle), and profile (right)

4 METHODS

4.1 Recruitment & Participants

For Phase 1 of our study, we recruited older adults from three under-resourced community sites located in a southern U.S. city: a suburban church (n=7) and two publicly subsidized living facilities for older adults (n=5). Four were ages 55-64 and eight were over 65. Ten identified as female and two identified as male. Eight self-identified as having diabetes, and four as being caregivers of someone with diabetes.

For Phase 2, we recruited a convenience sample of undergraduate (n=8) and graduate students (n=2) who knew someone with diabetes. The students were all under 30 years old and enrolled at a university in the same southern U.S. city. Eight of the students identified as female and two identified as male. Two of the students study engineering and eight study computer science or an adjacent field.

To be included in the study, participants had to self-report a diagnosis of Type 2 Diabetes or know someone diagnosed with diabetes. The study was IRB-approved and all participants provided written consent. Study sessions were audio recorded in support of analysis. Participants received \$20 for their participation in the form of a gift card.

4.2 Study Design & Analysis

Sessions with the older adults were conducted at their recruitment community site (suburban church or living facility) and sessions with students were conducted at their university. We first conducted a cognitive walkthrough [37] to see which components of our application are confusing to users and/or signal lower levels of familiarity with technology. Prior to beginning the walkthrough, participants were informed that the research team is "developing

an application for individuals with diabetes who may have diabetic foot ulcers.” The walkthrough included a task for each major component of the app: image capture, video capture, vibration test, walking test, reviewing previous patient recordings, and locating the patient’s profile information. The interviewer explained the components of the application and their purpose as needed. At the end of the study, via a semi-structured interview, participants were asked to explain what they understood regarding the app features. They were asked if they had any further questions about the app and what additional features would enhance their user experience.

Our qualitative data analysis focused on understanding the patterns shown by community members and students, and their experiences using the app. Bottom-up thematic analysis [7, 8] was performed on transcripts from the 22 cognitive walkthroughs and interviews. The 1st and 2nd authors coded the transcripts and engaged in a reflexive discussion to identify and agree upon themes.

5 FINDINGS

Four key themes were identified within the transcripts: technology perception, app navigation and technology skill, connecting the app to diabetes, and recommended app changes. Below we refer to the 12 older adults as A1-12 and the 10 university students as S1-10.

5.1 Technology Perception

While university students expressed no hesitation before starting the cognitive walkthroughs, the older adults seemed to have lower confidence in themselves before trying the application and as they used it. A11 expressed frustration four times in their interview, saying that they are not very “tech savvy” and laughing at themselves when they could not figure out a feature. A10 reacted similarly, expressing concern twice in their interview, when they had difficulty taking a photo. When A5 struggled with completing a task, they said, “You know, because I’m not familiar with doing all that...” showing that they do not have much experience with applications in general. Similarly, when asked about telemedicine, A3 said, “I’m old-fashioned honey, I’m almost 80 years old and I must see the doctor in person. Yes, I must. I don’t like that new technology,” showing discomfort with technology in the medical field.

5.2 App Navigation and Technology Skill

App navigation refers to understanding the application’s three main icons (clipboard, plus sign, profile icon) when prompted (See Figure 1). Table 1 shows if participants were able to successfully complete the task, which includes navigating to the correct part of the app.

Older adults had difficulty understanding the icons within the application, which inhibited their ability to complete the given task. Only two (17%) of the older adults found and used all icons correctly without assistance. In particular, participants did not intuitively understand that the plus button was to add a new record. When asked to navigate to the profile details, A8 expressed, “I would think it is this, that little person there. But I don’t know that someone who is not... if you’re looking at an older person who is not really used to using... [...] but I myself, I would get that.”

University students understood how to navigate the app but expressed concern for those with less tech literacy. Nine out of ten understood how to use all the icons without assistance. S7 clicked

Table 1: Cognitive Walkthrough Task Success Rates

Walkthrough Task		Older Adults (N=12)	Students (N=10)
Image Capture		83%	100%
Video Capture		75%	100%
Vibration Test		100%	100%
Gait Test		100%	100%
Icon Recognition	Clipboard	42%	100%
	Plus Sign	16%	90%
	Profile	75%	90%

the wrong icon (profile) when asked to find and record the first test (plus sign), but had stated that they were not familiar with apps and interfaces because they had not formally studied computer science. All of the students suggested changing the plus button as it would be difficult for those who are not tech savvy to understand.

Both university students and older adults needed guidance in navigating the application, and the plus button was difficult to comprehend. However, the university students had more success in walkthrough completion. Our findings echo previous work [38] that older adults may have knowledge gaps in icon comprehension.

5.3 Connecting the App to DFU Knowledge

While older adults understood the overarching meaning of each walkthrough task, they had difficulty understanding verbiage from the app and connecting it to the associated aspect of diabetic foot health. For example, A4 understood that the walking test was to track their walking, but had difficulty understanding that tracking gait could be related to the development of DFUs. The second most questioned feature was the vibration test (A3, A4, A9), but once explained, they had a better idea of how sensation was linked to nerve damage. A6 specifically mentioned that the vibration feature was “unique.” A3 expressed curiosity after they understood the vibration test and potential signs of neuropathy and asked if they could recreate the vibration test on their personal device.

Although the students have family members impacted by diabetes, they demonstrated limited diabetes knowledge and asked for more clarification on the app’s connection to diabetes. While seven students (70%) knew about DFUs, none were familiar with the capillary refill test, four (40%) knew about neuropathy in those with diabetes, and four understood the purpose of the walking test.

Both the older adults and students struggled to connect their diabetes knowledge to the app. While the students were thinking of a family member, older adults were speaking mostly from first-hand experience. Our findings suggest that mHealth app design should include information and guidance from the perspective of a user living with the associated chronic condition, as well as insights from those who may act as an informal caregiver. Although our app is usable for university students, its utility or usefulness may also be reliant on delivering health knowledge to its users.

5.4 Recommended Application Changes

The older adults suggested that an introductory walkthrough would enhance their understanding of the app. A4 expressed frustration and said, "I don't know how to do this..." when asked to use the video capture, but completed the task once given some assistance. To this point, A7 said, "I think once the person has shown them where the things are on the screen, it would be very easy." A10 echoed this statement saying "you simplified it for me...I appreciate it," after the interviewer explained a feature. A6 noted that a manual in the beginning may help in understanding. Older adults also discussed the inclusion of colloquial language. A9 said "what might be a concern would be an educational level," suggesting the application read "take a picture" instead of "image capture," because the latter "are not usual words."

University students suggested multiple user interface design changes to make the application more user-friendly with easier flow. S3 noted that "at first you don't realize that it's a scroll interface" and suggested making the application more user-friendly by arranging the four records in a grid on the "Add a Record" page. S6 also echoed this thought. S2, S3, and S7 stated that the clipboard icon made it seem like there was a task list to accomplish, rather than the actual purpose of viewing previous recordings. S2, S3, and S6 emphasized having a more aesthetic design with a larger font size, especially for those with visual impairments. S9 also said that having an audio feature for those who may not be able to read or those with declining eyesight would help them understand the features.

The students gave more design feedback as compared to older adults, who spoke more of difficulty in comprehension. Older adults specifically expressed that although the design may not be immediately intuitive, if given guidance prior to using the application, they were confident they could learn how to use it. In contrast, the students felt it was vital that the aspects of the app were easy to understand at first glance, and expressed concern that those who are visually impaired may struggle with app usability.

6 DESIGN IMPLICATIONS

6.1 Accessible Technology & Diabetes Education

Our study has multiple implications for accessible mHealth design. We echo previous findings that icon knowledge may be limited for older adults [38], but found that they were optimistic about their ability to successfully navigate the app if provided a walkthrough early on. In addition, having a larger font size and a text-to-speech feature would assist those who may be visually impaired. For older adults living with diabetes, this may be a widespread concern due to normal eyesight decline and diabetic retinopathy [6].

Our study also found that while most older adults were able to use their phones to complete walkthrough tasks (taking pictures, videos, etc.), the utility of the mHealth app was limited by participants' familiarity with key aspects of diabetic foot health. Previous research has shown that patient-centered diabetes education is crucial in managing symptoms and mitigating disease progression [32]. Educational material with simple language, potentially alongside the app, may prove to be beneficial in educating users [39]. Finally, to be cognizant of varying education levels, using more common language rather than technical language in the application would be beneficial for users.

6.2 Including Older Adults in Design

Older adults are often not represented in technology design: their lower understanding is not considered as valuable in the design field [25]. However, our study suggests that older adults can provide valuable feedback to make applications more accessible. Community-based design and research has been shown to provide a platform for individuals from lower socioeconomic backgrounds to express their perspectives [17]. This approach facilitates discussions on health matters within their community, fostering a sense of inclusion and belonging. Moreover, it enables researchers to gain insights into the social, cultural, and economic factors shaping participants' viewpoints [16, 17]. When users understand how and why a system behaves in a certain way, the system becomes more useful and effective for them [9]. This has been echoed by our participants: they emphasized that once they understood the application, they appreciated the potential value that it could bring to their lives.

6.3 Reflecting on the Digital Divide

Our study suggests that older adults living with diabetes within under-resourced communities face challenges in understanding technology and the utility of mHealth applications. Community-based educational initiatives have proven beneficial in addressing this gap [28, 36]. Designers may consider implementing community-based technology strategies when teaching others about their applications, especially in under-resourced settings.

To explore this digital divide, it was important to also evaluate our application with those more familiar with technology, i.e., university students studying computer science or engineering. We found that almost all of the students understood the icons in the mobile application, but most older adults struggled with icon recognition. Through Phase 2 of the study, we were able to see how much of a difference technology-integrated education can have on the understanding of an app, and how some populations with lower levels of technology often have a lapse in this domain [36].

Our study suggests that older adults are able to utilize a smartphone to assist in health self-monitoring (e.g., through taking photos, videos). However, it is important to ensure that DFU technology is intuitive to its users. The design of our diabetic foot care mHealth application must be improved to better cater to its users, such as by improving the accessibility and by providing educational walkthroughs. Further, the utility of mobile health interventions may be limited by users' health knowledge. By advocating for community-based educational initiatives and including both participants with lower tech literacy and those who are very tech savvy within the design process, designers can develop more accessible mHealth applications to narrow the digital divide.

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